For a centain gas which deviates a little from ideal behaviour. A plot between P/ρ vs P was founds to be nonlinear, the intercept no y-axis will be :

(A) $\frac{RT}{M}$ ✓ (B) $\frac{M}{RT}$ (C) $\frac{MZ}{RT}$ (D) $\frac{R}{TM}$

Sol^n:

Ideal gas, $PM = \rho \times RT$

$$\boxed{\frac{P}{\rho} = \frac{RT}{M} + 0 \times P}$$

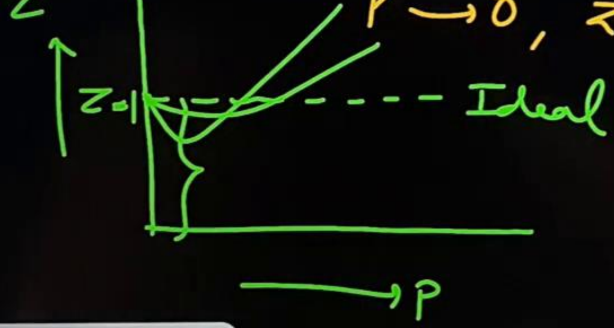
For non-ideal gas this slope $\neq 0$.

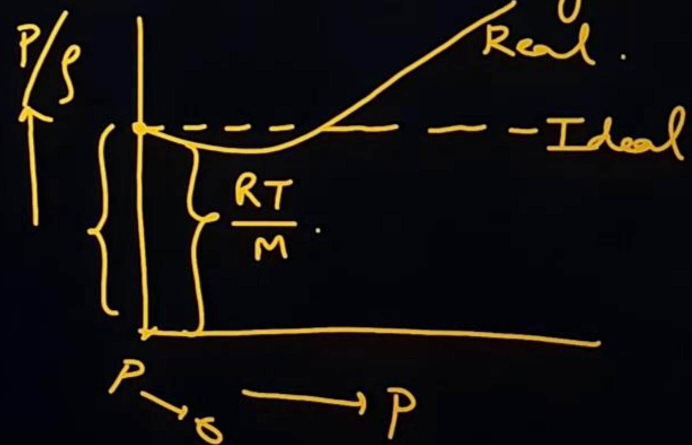
For an ideal gas.

Real gas, $PM = Z \times \rho RT$

$$\frac{P}{\rho} = Z\frac{RT}{M}.$$

$P \to 0$, $Z = 1 \Rightarrow \left(\frac{P}{\rho}\right)_{P\to 0} = \frac{RT}{M}$





18:29

What is the compressibility factor (Z) for 0.02 mole of a van der Waals' gas at pressure of 0.1 atm. Assume the size of gas molecules is negligible.

Given : RT = 20 L atm mol^{-1} and a = 1000 atm L^2 mol^{-2}

Sol^n: $Z = ?$, $n = 0.02$ mol, $P = 0.1$ atm, $b = 0$

$a = 1000$

$RT = 20$ L atm mol^{-1}.

$PV = Z \times nRT.$

$0.1 \times V = Z \times 0.02 \times 20$

$Z = \dfrac{0.1V}{0.4} = \dfrac{V}{4}$

$Z = \dfrac{2}{4} = 0.5$ Ans

$\left(P + \dfrac{an^2}{V^2}\right)(V - nb) = nRT.$

$\left(0.1 + \dfrac{(1000 \times 4 \times 10^{-4})}{V^2}\right) \times (V - 0) = 0.02 \times 20.$

$0.1V + \dfrac{0.4}{V} = 0.4.$

$V^2 - 4V + 4 = 0 \Rightarrow (V-2)^2 = 0$

$V = 2$ lt

18:35

If Pd v/s P(where P denotes pressure in atm and d denotes density in gm/L) is plotted for He gas (assume ideal) at a particular temperature. If $\left[\frac{d}{dP}(Pd)\right]_{P=8.21\,atm} = 5$, then the temperature will be

(A) 160 K (B) 320 K (C) 80 K (D) none of these

Sol^n: He : Ideal gas.

$PM = dRT$

$d = \frac{PM}{RT}$

$d \times PM = dRT \times d$

$Pd = \frac{RT}{M} \times d^2$

$= \frac{\cancel{RT}}{\cancel{M}} \times \frac{P^2 M^{\cancel{2}}}{R^{\cancel{2}} T^{\cancel{2}}} = \frac{P^2 M}{RT}$

or $P \times PM = dRT \times P$

$Pd = \frac{M}{RT} \times P^2$

$\frac{d(Pd)}{dP} = \frac{M}{RT} \times 2P$

Pd vs P

$\left\{\frac{d(Pd)}{dP}\right\}_{P=8.21atm} = 5$

$\frac{M}{RT} \times 2P = 5$

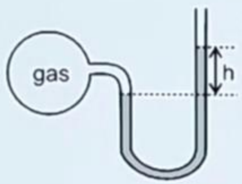
$\frac{4}{0.\cancel{0821} \times T} \times 2 \times \cancel{8.21}^{100} = 5$

$\Rightarrow T = \frac{400 \times 2}{5}$

$= 160K$

18:42

A bulb of constant volume is attached to a manometer tube open at other end as shown in figure. The manometer is filled with a liquid of density (1/3)[rd] that of mercury. Initially h was 228 cm. Through a small hole in the bulb gas leaked assuming pressure decreases as $\frac{dp}{dt} = -kP$. If value of h is 114 cm after 14 minutes. What is the value of k (in hour–1) ?



(A) 0.6 (B) 1.2 (C) 2.4 (D) none of these

$h_1 d_1 = h_2 d_2$

$228 \times \frac{1}{3} d_{Hg} = h_{Hg} \times d_{Hg}$

$h_{Hg} = 76$ cm of Hg
$= 1$ atm.

Sol^n: $V \longrightarrow$ constant , $\frac{dP}{dt} = -kP$ [Assuming $P_{atm} = 1$ atm).

$d_{(l)} = \frac{1}{3} \times d_{Hg(l)}$

$P_i = \underbrace{P_{atm}} + \underbrace{h\,dg}$
$= 1 + 1 = 2$ atm.

$P_f = P_{atm} + 114 \times \frac{1}{3}$
$= (1 + 0.5)$ atm $= 1.5$ atm.

$\int_{P_i}^{P_f} \frac{dP}{P} = -k \int_0^t dt$

$\ln \frac{P_f}{P_i} = -k \times t$

$2.303 \times \log \frac{1.5}{2} = -k \times \left(\frac{14}{60}\right)$.

$k = \frac{60}{14} \times 2.303 \times \log \frac{4}{3}$
$= \frac{60}{14} \times 2.3 \times (0.6 - 0.48)$
$= \frac{60}{14} \times 2.3 \times 0.12$ hr^{-1}
$= 1.2$ hr^{-1}.

18:50

Vander Waal's gas equation can be reduced to virial equation and virial equation (in terms of volume) is

$$Z = A + \frac{B}{V_m} + \frac{C}{V_m^2} + \ldots\ldots$$

where A = first virial coefficient, B = second virial coefficient, C = third virial coefficient. The third virial coefficient of Hg(g) is 625 $(cm^2/mol)^2$. What volume is available for movement of 10 moles He(g) atoms present in 50 L vessel?

(A) 49.75 L (B) 49.25 L (C) 25 L (D) 50 L

He(g)

$(cm^3/mol) \rightarrow$ unit of 'b'

$PV = Z \times nRT$ or $PV_m = ZRT$

$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$

$P = \left(\frac{RT}{V_m - b} - \frac{a}{V_m^2}\right) \times \frac{V_m}{RT}$

$\times \frac{V_m}{RT}$

$Z = \frac{V_m}{V_m - b} - \frac{a}{RTV_m}$

$= \frac{1}{\left(1 - \frac{b}{V_m}\right)} - \frac{a}{RTV_m}$

$= \left(1 - \frac{b}{V_m}\right)^{-1} - \frac{a}{RTV_m}$

$= 1 + \frac{b}{V_m} + \frac{b^2}{V_m^2} + \frac{b^3}{V_m^3} + \ldots - \frac{a}{RTV_m}$

Solⁿ: $Z = A + \frac{B}{V_m} + \frac{C}{V_m^2} + \ldots$

Imp

$Z = 1 + \frac{(b - a/RT)}{V_m} + \frac{b^2}{V_m^2} + \frac{b^3}{V_m^3} + \ldots$ (Virial eqⁿ of state)

$C = b^2 = 625 \Rightarrow b = 25\ cm^3/mol$

$V = 50$ lit, $= 0.025$ lit/mol

Free volume $= (V - nb) = 50 - 10 \times 0.025 = 50 - 0.25$

$= 49.75$ lit

19:05

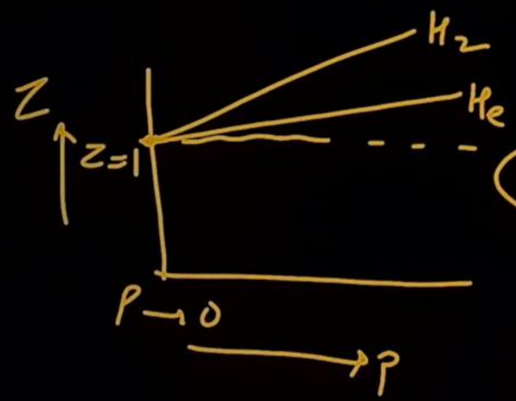
Match gases under specified conditions listed in Column I with their properties in Column II.

Column-I		Column-II
(A) Hydrogen gas (P = 200 atm, T = 273 K)	$\rightarrow p, s$	(p) Compressibility factor $\neq 1$
(B) Hydrogen gas (P ~ 0, T = 273 K)	$\rightarrow r$	(q) Attractive forces are dominant
(C) CO_2 (P = 1 atm, T = 273 K)	$\rightarrow p, q$	(r) PV = nRT
(D) Real gas with very large molar volume	$\rightarrow r$	(s) P(V – nb) = nRT

JEE-Adv

Sol^n: (A) H_2 gas ($P = 200$ atm → high pr, $T = 273K$, $Z > 1$)

(Repulsive forces.



$A \rightarrow p, s$

$P(V-nb) = nRT.$

$\left(P + \cancel{\frac{an^2}{V^2}}\right)(V-nb) = nRT$

a = very small for He and H_2

(B) H_2 gas, $P \cong 0$ (ideal behaviour)

$B \rightarrow r$

(C) CO_2, 1 atm, 273 K

attractive forces dominant

$Z \neq 1$

$C \rightarrow p, q$

$D \rightarrow r$

(D) $V_m \rightarrow \infty$, $P \rightarrow 0$, $Z \rightarrow 1$

ideal behaviour,

19:13

One mole of a gas changed from its initial state (15 lt, 2 atm) to final state (4 lt,10 atm) reversibly. If this change can be represented by a straight line in P–V curve maximum temperature (approximate), the gas attained is x × 10^2 K. Then find the volue of x.

Sol^n: $n = 1$ mol, (15 lit, 2 atm ⟶ 4 lit, 10 atm)

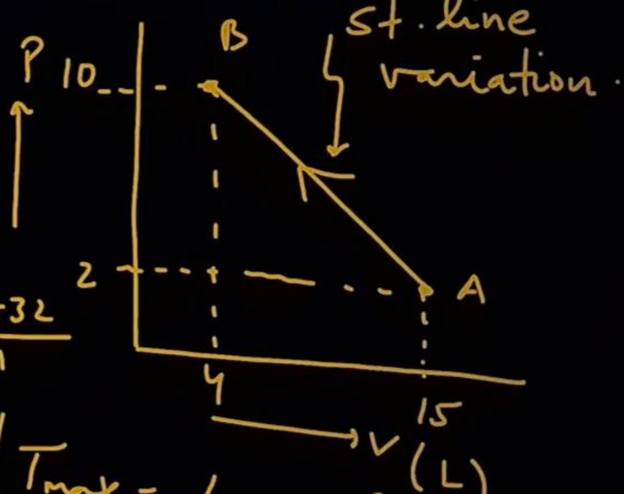
$y = mx + C \Rightarrow P = m \times V + C$

$2 = 15m + C$

$10 = 4m + C$ $\Rightarrow 11m = -8$

$m = -\frac{8}{11}$

$C = 10 + 4 \times \frac{8}{11} = \frac{110 + 32}{11} = \frac{142}{11}$



$P = -\frac{8V}{11} + \frac{142}{11}$

$PV = nRT$

$PV = 1 \times RT \Rightarrow$

$\frac{RT}{V} = (-8V + 142) \times \frac{1}{11}$

$T = \frac{1}{11R}\left[-8V^2 + 142V\right]$

For maxima of temp, $\frac{dT}{dV} = 0$.

$\frac{1}{11R}\left[-16V + 142\right] = 0 \Rightarrow V = \frac{142}{16} = \frac{71}{8}$ lit

for max. temp.

$T_{max} = \frac{1}{11 \times 0.0821}\left[-16 \times \frac{71}{8} \times \frac{71}{8} + 142 \times \frac{71}{8}\right]$

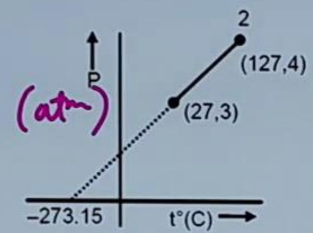
$\simeq 700$ K

$x \times 10^2 = 700$ K

$x = 7$ Ans.

19:18

Two moles of an ideal gas udergoes the following process. Given that $\left(\frac{\partial P}{\partial T}\right)_V$ is $x \times 10^{-y}$, then calculate the volue of $(x + y)$



$x + y = 3$

Ans

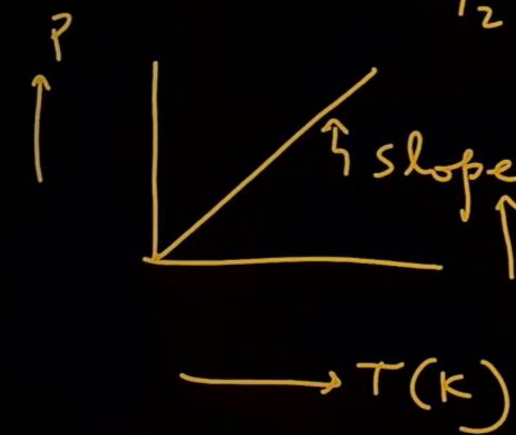
Sol^n :

$T_1 = 27 + 273 = 300\,K$, $P_1 = 3$ atm

$T_2 = 127 + 273 = 400\,K$, $P_2 = 4$ atm

$\frac{P_1}{T_1} = \frac{P_2}{T_2} \Rightarrow \frac{3}{300} = \frac{4}{400}$

So volume is constant during the process.



slope $= \frac{nR}{V}$

$PV = nRT$

$P = \left(\frac{nR}{V}\right) \times T$

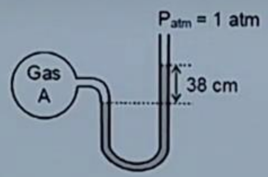
$\left(\frac{dP}{dT}\right)_V = x \times 10^{-y} = \frac{4-3}{400-300} = \frac{1}{100} = 1 \times 10^{-2}$

$\cancel{\frac{nR}{V} = x \times 10^{-y} \Rightarrow \frac{2 \times 0.0821}{}}$

$x = 1$, $y = 2$, $(x+y) = 3$

Ans

A manometer contains a liquid of density 5.44g/cm³ is attached to a flask containing gas 'A' as follows



... undergoes 30% trimerisation [$3A(g) \rightleftharpoons A_3(g)$] then the difference in height of the ... in two columns is :

(B) 7.6 cm (C) 3.04 cm (D) 15.2 cm

$(p_A)_i = P_{atm} + h d g$

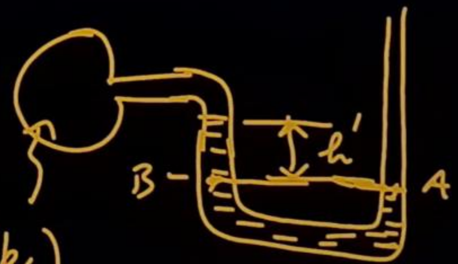
$= (1 + 0.2)$ atm $= 1.2$ atm.

$(g) \rightleftharpoons A_3(g)$

-0.3×1.2 $\quad \frac{1}{3} \times (0.3 \times 1.2)$

$0.36 + 0.12$

$- 0.24 = 0.96$ atm



$(p_f)_A = 0.96$ atm

1 atm $= h' + 0.96$.

$h' = 0.04 \times 76$ cm of Hg

$h_1 d_1 = h_2 d_2$

$38 \times 5.44 = h_{Hg(l)} \times 13.6$

$h_{Hg(l)} = \frac{38 \times 5.44}{13.6 \times 76}$ atm

$= 0.2$ atm

h' in terms of given liquid $= \frac{0.04 \times 76}{0.4}$

$= 7.6$ cm of liquid

Q.1

19:35

19:40

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT.$$

(a) <u>At low pr</u>: Intermolecular forces are dominant

$\cancel{b}$ ignored relative V_m.

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT.$$

$$Z = 1 - \frac{a}{RTV_m}, \quad Z < 1$$

(b) <u>High pr</u>: Repulsive forces are dominant

$\frac{a}{V_m^2}$ is ignored relative 'p'

$$P(V_m - b) = RT.$$

$$Z = 1 + \frac{Pb}{RT}, \quad Z > 1$$

(c) $\underbrace{H_2 \ \& \ He}$

$\hookrightarrow a \rightarrow$ very low value.

$$Z = 1 + \frac{Pb}{RT}, \quad Z > 1$$

when T is Room temp or 0°C or in general $T > T_b$

$$T_b = \frac{a}{Rb}$$

(D) when $P \rightarrow 0$ or $V_m \rightarrow \infty$.

$\therefore Z \rightarrow 1$ (ideal)

$PV = nRT$